r

Habet Madoyan

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Create 2 new vectors

```
revenue <- c(140, 50, 20, 120, 240)
expenses <- c(24, 50, 100, 50, 10)
```

```
names(revenue)<-c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday")
```

Subseting vector by element name

```
revenue["Monday"]
```

```
## Monday
##    140
```

Subseting element from vector by position. In R indexing starts from 1

```
revenue[4]
```

```
## Thursday
##      120
```

You can select 1 and 3 elements

```
revenue[c(1,3)]
```

```
##    Monday Wednesday
##       140        20
```

Or elements from 1 to 4

```
1:4
```

```
## [1] 1 2 3 4
```

```
revenue[c(1:4)]
```

```
##    Monday   Tuesday Wednesday  Thursday
##       140        50        20       120
```

Find daily income

```
inc_d<-revenue-expenses
inc_d
```

```
##    Monday   Tuesday Wednesday  Thursday    Friday
##       116         0       -80        70       230
```

Find the days with positive revenue

```
# This will give you a boolean vector
inc_d>0
```

```
##    Monday   Tuesday Wednesday  Thursday    Friday
##      TRUE     FALSE     FALSE      TRUE      TRUE
```

subset

```
inc_pos<-inc_d[inc_d>0]
inc_pos
```

```
##   Monday Thursday   Friday
##      116       70      230
```

```
#Find the non-negative income days
# find days when income was 0
```

You can make comparison between vectors two

```
revenue>expenses
```

```
##    Monday   Tuesday Wednesday  Thursday    Friday
##      TRUE     FALSE     FALSE      TRUE      TRUE
```

Recycling

```
u<-c(10,20,30)
v<-c(1,2,3,4,5,6,7,8,9)
u+v
```

```
## [1] 11 22 33 14 25 36 17 28 39
```

You can use sum() command to get the sum of all element in the vector

```
sum(inc_d)
```

```
## [1] 336
```

finding the average

```
mean(inc_d)
```

```
## [1] 67.2
```

you can do other operations with vectors too

```
# multiply by a number
inc_d*2
```

```
##    Monday   Tuesday Wednesday  Thursday    Friday
##       232         0      -160       140       460
```

dot product of the vectors

```
a<-c(1,4,5,3)
b<-c(5,3,9,8)
a%*%b
```

```
##      [,1]
## [1,]   86
```

Multiplaction of two vectors

```
a%*%t(b)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    5    3    9    8
## [2,]   20   12   36   32
## [3,]   25   15   45   40
## [4,]   15    9   27   24
```

```r
d<-c(0,1,6,NA,5 ,NaN)
```

```r
is.na(d)
```

```
## [1] FALSE FALSE FALSE  TRUE FALSE  TRUE
```

```r
!is.na(d)
```

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE FALSE
```

MAtrix

```r
matrix(1:9, byrow = TRUE, nrow = 3)
```

```
##      [,1] [,2] [,3]
## [1,]    1    2    3
## [2,]    4    5    6
## [3,]    7    8    9
```

```r
# Box office Star Wars (in millions!)
new_hope <- c(460.998, 314.4)
empire_strikes <- c(290.475, 247.900)
return_jedi <- c(309.306, 165.8)

# Create box_office
box_office <-c(new_hope, empire_strikes, return_jedi)
```

Construct star_wars_matrix

```r
star_wars_matrix <- matrix(box_office, nrow = 3, byrow = TRUE)
star_wars_matrix
```

```
##         [,1]  [,2]
## [1,] 460.998 314.4
## [2,] 290.475 247.9
## [3,] 309.306 165.8
```

We could have done the same using rbind command

```r
star_wars_m1<-rbind(new_hope, empire_strikes, return_jedi)
star_wars_m1
```

```
##                   [,1]  [,2]
## new_hope       460.998 314.4
## empire_strikes 290.475 247.9
## return_jedi    309.306 165.8
```

naming rows and columns

```r
region <- c("US", "non-US")
titles <- c("A New Hope", "The Empire Strikes Back", "Return of the Jedi")
```

```
# Name the columns with region
colnames(star_wars_matrix) <- region

# Name the rows with titles
rownames(star_wars_matrix) <- titles

# Print out star_wars_matrix
star_wars_matrix
```

```
##                             US non-US
## A New Hope              460.998  314.4
## The Empire Strikes Back 290.475  247.9
## Return of the Jedi      309.306  165.8
```

Lets get worldwide box-office for the tree movies

```
worldwide_vector <- rowSums(star_wars_matrix)
worldwide_vector
```

```
##              A New Hope The Empire Strikes Back      Return of the Jedi
##                 775.398                 538.375                 475.106
```

Get US and Non-US box office for all tree movies combined

```
combined_vector<-colSums(star_wars_matrix)
combined_vector
```

```
##       US   non-US
## 1060.779  728.100
```

```
# take the first row -  A new hope
new_hope<-star_wars_matrix[1,]
new_hope
```

```
##      US  non-US
## 460.998 314.400
```

```
# take the first column
US<-star_wars_matrix[,1]
```

```
#US box office for "A new hope"
star_wars_matrix[1,1]
```

```
## [1] 460.998
```

```
# You can do the subseting by the names of rows and columns as well,
# if they are specified
star_wars_matrix[,"non-US"]
```

```
##              A New Hope The Empire Strikes Back      Return of the Jedi
##                   314.4                   247.9                   165.8
```

```
star_wars_matrix["A New Hope","non-US"]
```

```
## [1] 314.4
```

```
data(mtcars)
# ?mtcars
# look at the top 5 observations
head(mtcars, n=5)
```

```
##                    mpg cyl disp  hp drat    wt  qsec vs am gear carb
## Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
## Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
## Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
## Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
## Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
```

Look at the bottom file observations

```
tail(mtcars, n=5)
```

```
##                 mpg cyl  disp  hp drat    wt qsec vs am gear carb
## Lotus Europa   30.4   4  95.1 113 3.77 1.513 16.9  1  1    5    2
## Ford Pantera L 15.8   8 351.0 264 4.22 3.170 14.5  0  1    5    4
## Ferrari Dino   19.7   6 145.0 175 3.62 2.770 15.5  0  1    5    6
## Maserati Bora  15.0   8 301.0 335 3.54 3.570 14.6  0  1    5    8
## Volvo 142E     21.4   4 121.0 109 4.11 2.780 18.6  1  1    4    2
```

```
str(mtcars)
```

```
## 'data.frame':    32 obs. of  11 variables:
##  $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
##  $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
##  $ disp: num  160 160 108 258 360 ...
##  $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
##  $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
##  $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
##  $ qsec: num  16.5 17 18.6 19.4 17 ...
##  $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
##  $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
##  $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
##  $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

look at the summary of the dataframe

```
summary(mtcars)
```

```
##       mpg             cyl             disp             hp
##  Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0
##  1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5
##  Median :19.20   Median :6.000   Median :196.3   Median :123.0
##  Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7
##  3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0
##  Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0
##       drat             wt             qsec             vs
##  Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000
##  1st Qu.:3.080   1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000
##  Median :3.695   Median :3.325   Median :17.71   Median :0.0000
##  Mean   :3.597   Mean   :3.217   Mean   :17.85   Mean   :0.4375
##  3rd Qu.:3.920   3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000
##  Max.   :4.930   Max.   :5.424   Max.   :22.90   Max.   :1.0000
##        am              gear            carb
##  Min.   :0.0000   Min.   :3.000   Min.   :1.000
##  1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000
##  Median :0.0000   Median :4.000   Median :2.000
##  Mean   :0.4062   Mean   :3.688   Mean   :2.812
##  3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000
##  Max.   :1.0000   Max.   :5.000   Max.   :8.000
```

Subseting the dataframes

```
# get the dimensions of the dataframe
dim(mtcars)
```

```
## [1] 32 11
```

Create new dataframe with the first two columns and all the rows

```
df1<-mtcars[,1:2]
dim(df1)
```

```
## [1] 32  2
```

Create new dataframe with column 1,2,3 and 6, and first 10 rows

```
df2<-mtcars[1:10,c(1:3,6)]
dim(df2)
```

```
## [1] 10  4
```

Create new dataframe with all the variables, except 1 and 4

```
df3<-mtcars[,-c(1,4)]
dim(df3)
```

```
## [1] 32  9
```

if the columns or rows have names, you can subset by names as well

```
df4<-mtcars[,c("mpg", "gear", "disp")]
names(df4)
```

```
## [1] "mpg"  "gear" "disp"
```

you can access a column in dataframe using $ sign if columns have names

```
mpg<-mtcars$mpg
mpg
```

```
##  [1] 21.0 21.0 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 17.8 16.4 17.3 15.2
## [15] 10.4 10.4 14.7 32.4 30.4 33.9 21.5 15.5 15.2 13.3 19.2 27.3 26.0 30.4
## [29] 15.8 19.7 15.0 21.4
```

You can subset dataframe by rows by condition on any column

```
# keep only rows where mpg is more than or equal to 15
df5<-mtcars[mtcars$mpg>=15,]
summary(df5)
```

```
##       mpg             cyl             disp             hp       
##  Min.   :15.00   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
##  1st Qu.:17.55   1st Qu.:4.000   1st Qu.:120.2   1st Qu.: 94.0  
##  Median :21.00   Median :6.000   Median :160.0   Median :110.0  
##  Mean   :21.47   Mean   :5.852   Mean   :196.3   Mean   :131.6  
##  3rd Qu.:23.60   3rd Qu.:8.000   3rd Qu.:275.8   3rd Qu.:175.0  
##  Max.   :33.90   Max.   :8.000   Max.   :400.0   Max.   :335.0  
##       drat             wt             qsec             vs        
##  Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000  
##  1st Qu.:3.115   1st Qu.:2.393   1st Qu.:16.96   1st Qu.:0.0000  
##  Median :3.770   Median :3.170   Median :18.00   Median :1.0000  
##  Mean   :3.666   Mean   :2.945   Mean   :18.03   Mean   :0.5185  
##  3rd Qu.:4.000   3rd Qu.:3.450   3rd Qu.:19.17   3rd Qu.:1.0000  
##  Max.   :4.930   Max.   :4.070   Max.   :22.90   Max.   :1.0000  
##        am              gear            carb      
##  Min.   :0.0000   Min.   :3.000   Min.   :1.000  
##  1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:1.500  
##  Median :0.0000   Median :4.000   Median :2.000  
##  Mean   :0.4815   Mean   :3.815   Mean   :2.593  
##  3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:3.500  
##  Max.   :1.0000   Max.   :5.000   Max.   :8.000
```

Can subset by rwos using more than 1 condition

```
# AND logical operator
df6<-mtcars[mtcars$mpg>15 & mtcars$carb<3,]
summary(df6)
```

```
##       mpg             cyl             disp             hp       
##  Min.   :15.20   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
##  1st Qu.:19.20   1st Qu.:4.000   1st Qu.: 95.1   1st Qu.: 66.0  
##  Median :22.80   Median :4.000   Median :121.0   Median : 97.0  
##  Mean   :23.61   Mean   :5.176   Mean   :177.7   Mean   :104.4  
##  3rd Qu.:27.30   3rd Qu.:6.000   3rd Qu.:258.0   3rd Qu.:113.0  
##  Max.   :33.90   Max.   :8.000   Max.   :400.0   Max.   :175.0  
##       drat             wt             qsec             vs        
##  Min.   :2.760   Min.   :1.513   Min.   :16.70   Min.   :0.0000  
##  1st Qu.:3.150   1st Qu.:2.140   1st Qu.:17.05   1st Qu.:0.0000  
##  Median :3.770   Median :2.780   Median :18.61   Median :1.0000  
##  Mean   :3.692   Mean   :2.709   Mean   :18.73   Mean   :0.7059  
##  3rd Qu.:4.080   3rd Qu.:3.435   3rd Qu.:19.90   3rd Qu.:1.0000  
##  Max.   :4.930   Max.   :3.845   Max.   :22.90   Max.   :1.0000  
##        am              gear            carb      
##  Min.   :0.0000   Min.   :3.000   Min.   :1.000  
##  1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:1.000  
##  Median :0.0000   Median :4.000   Median :2.000  
##  Mean   :0.4706   Mean   :3.706   Mean   :1.588  
##  3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:2.000  
##  Max.   :1.0000   Max.   :5.000   Max.   :2.000
```

```r
# The OR logical operator
df7<-mtcars[mtcars$mpg>15 | mtcars$carb<3,]
summary(df7)
```

```
##       mpg             cyl             disp             hp       
##  Min.   :15.20   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
##  1st Qu.:17.88   1st Qu.:4.000   1st Qu.:120.2   1st Qu.: 93.5  
##  Median :21.00   Median :6.000   Median :160.0   Median :110.0  
##  Mean   :21.72   Mean   :5.769   Mean   :192.3   Mean   :123.8  
##  3rd Qu.:24.00   3rd Qu.:8.000   3rd Qu.:275.8   3rd Qu.:168.8  
##  Max.   :33.90   Max.   :8.000   Max.   :400.0   Max.   :264.0  
##       drat             wt             qsec             vs        
##  Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000  
##  1st Qu.:3.098   1st Qu.:2.356   1st Qu.:17.02   1st Qu.:0.0000  
##  Median :3.810   Median :3.160   Median :18.15   Median :1.0000  
##  Mean   :3.671   Mean   :2.921   Mean   :18.16   Mean   :0.5385  
##  3rd Qu.:4.040   3rd Qu.:3.440   3rd Qu.:19.30   3rd Qu.:1.0000  
##  Max.   :4.930   Max.   :4.070   Max.   :22.90   Max.   :1.0000  
##        am              gear            carb      
##  Min.   :0.0000   Min.   :3.000   Min.   :1.000  
##  1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:1.250  
##  Median :0.0000   Median :4.000   Median :2.000  
##  Mean   :0.4615   Mean   :3.769   Mean   :2.385  
##  3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:3.000  
##  Max.   :1.0000   Max.   :5.000   Max.   :6.000
```

Creating the final season schedule for Serie A.

The following varaibles describe the first 5 places for Serie A at 1996.

```
Team<-c("AC Milan","Juventus","Lazio","ACF Fiorentina","AS Roma","Parma AC")
Wins<-c(21,19,17,17,16,16)
Draws<-c(10,8,8,8,10,10)
Points<-c(73,65,59,59,58,58)
```

we can combine these variables into a dataframe

```
seriea<-data.frame(Team,Points, Wins, Draws)
class(seriea)
```

```
## [1] "data.frame"
```

Subseting the dataframes

```
mean(seriea$Points)
```

```
## [1] 62
```

```
mean(seriea[,"Points"])
```

```
## [1] 62
```

We can create a new variable within the dataframe that will show the number of games each team has lost (total 34 games were played)

```
seriea$Lost<-34-(seriea$Wins+seriea$Draws)
```

List

```
# Vector with numerics from 1 up to 10
my_vector <- 1:10

# Matrix with numerics from 1 up to 9
my_matrix <- matrix(1:9, ncol = 3)

# First 5 elements of the built-in data frame mtcars
my_df <- mtcars[1:5,]

# a character
my_char<-"DataMata"

# combine into a list
my_list<-list(my_vector, my_matrix, my_df, my_char)
my_list
```

```
## [[1]]
##  [1]  1  2  3  4  5  6  7  8  9 10
##
## [[2]]
##      [,1] [,2] [,3]
## [1,]    1    4    7
## [2,]    2    5    8
## [3,]    3    6    9
##
## [[3]]
##                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
## Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
## Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
## Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
## Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
## Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
##
## [[4]]
## [1] "DataMata"
```

```r
# Vector with numerics from 1 up to 10
my_vector <- 1:10

# Matrix with numerics from 1 up to 9
my_matrix <- matrix(1:9, ncol = 3)

# First 10 elements of the built-in data frame mtcars
my_df <- mtcars[1:5,]

# a character
my_char<-"DataMata"

# Adapt list() call to give the components names
my_list <- list(vec = my_vector, mat = my_matrix,
                df = my_df, char=my_char)

# Print out my_list
my_list
```

```
## $vec
##  [1]  1  2  3  4  5  6  7  8  9 10
##
## $mat
##      [,1] [,2] [,3]
## [1,]    1    4    7
## [2,]    2    5    8
## [3,]    3    6    9
##
## $df
##                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
## Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
## Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
## Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
## Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
## Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
##
## $char
## [1] "DataMata"
```

New list for movie shining

```
moviename<-"The Shining"
actors<-c("Jack Nicholson","Shelley Duvall","Danny Lloyd",      "Scatman Crothers","Barry Nelson" )
reviews<-data.frame(scores=c(4.5,4,5),
        sources=c("IMDb1", "IMDb2", "IMDb3"),
        comments =c("Best Horror Film I Have Ever Seen",
    "A truly brilliant and scary film from Stanley Kubrick", "A masterpiece of psychological horror"))
shining_list<-list(moviename=moviename, actors=actors, reviews=reviews)
shining_list
```

```
## $moviename
## [1] "The Shining"
##
## $actors
## [1] "Jack Nicholson"   "Shelley Duvall"   "Danny Lloyd"
## [4] "Scatman Crothers" "Barry Nelson"
##
## $reviews
##   scores sources                                          comments
## 1    4.5   IMDb1                      Best Horror Film I Have Ever Seen
## 2    4.0   IMDb2 A truly brilliant and scary film from Stanley Kubrick
## 3    5.0   IMDb3                 A masterpiece of psychological horror
```

Sub setting the list

One way to select a component is using the numbered position of that component. For example, to "grab" the first component of shining_list you type

```
shining_list[[1]]
```

```
## [1] "The Shining"
```

A quick way to check this out is typing it in the console. Important to remember: to select elements from vectors, you use single square brackets: [ ]. Don't mix them up!

You can also refer to the names of the components, with [[ ]] or with the $ sign. Both will select the data frame representing the reviews:

```
shining_list[["reviews"]]
```

```
##   scores sources                                              comments
## 1    4.5   IMDb1                       Best Horror Film I Have Ever Seen
## 2    4.0   IMDb2 A truly brilliant and scary film from Stanley Kubrick
## 3    5.0   IMDb3                  A masterpiece of psychological horror
```

```
shining_list$reviews
```

```
##   scores sources                                              comments
## 1    4.5   IMDb1                       Best Horror Film I Have Ever Seen
## 2    4.0   IMDb2 A truly brilliant and scary film from Stanley Kubrick
## 3    5.0   IMDb3                  A masterpiece of psychological horror
```

Besides selecting components, you often need to select specific elements out of these components. For example, with shining_list[[2]][1] you select from the second component, actors (shining_list[[2]]), the first element ([1])

```
shining_list[[2]][1]
```

```
## [1] "Jack Nicholson"
```

Adding new value to the list

simply use c

```
shining_list<-c(shining_list, year=1980)
shining_list
```

```
## $moviename
## [1] "The Shining"
##
## $actors
## [1] "Jack Nicholson"   "Shelley Duvall"   "Danny Lloyd"
## [4] "Scatman Crothers" "Barry Nelson"
##
## $reviews
##   scores sources                                          comments
## 1    4.5   IMDb1                     Best Horror Film I Have Ever Seen
## 2    4.0   IMDb2 A truly brilliant and scary film from Stanley Kubrick
## 3    5.0   IMDb3                 A masterpiece of psychological horror
##
## $year
## [1] 1980
```

```r
gender <- c("Male", "Female", "Female", "Male", "Male")
gender
```

```
## [1] "Male"   "Female" "Female" "Male"   "Male"
```

Make it a vector

```r
# Convert gender_vector to a factor
gender_f <- factor(gender)
gender_f
```

```
## [1] Male   Female Female Male   Male
## Levels: Female Male
```

```r
gender<-c(0,1,0,0,0,1)
# Here 0 stands for Male, 1 Stands for Female
gender<-factor(gender, levels=c(0,1), labels=c("Male", "Female"))
levels(gender)
```

```
## [1] "Male"   "Female"
```